## Type MS · Construct or complete a results or frequency table

### Answer 1 of 6 · Question 4 part a.i · 4 marks

2025 JUN · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 4 (a) (i) | | <table><tr><th>Conc (sugar) in au</th><th>Percentage change in mass</th></tr><tr><td>0.00</td><td>8.0</td></tr><tr><td>0.25</td><td>4.0</td></tr><tr><td>0.50</td><td>-1.0</td></tr><tr><td>1.00</td><td>-7.0</td></tr><tr><td>1.50</td><td>-10.0</td></tr></table><br>• two row/column table;<br>• concentration (sugar) in au;<br>• percentage/% change in mass;<br>• 0.0 to 1.50/1.50-0.00; | Ignore units in the body of the table<br><br>Accept AU for MP2<br><br>Ignore % symbol for MP3<br><br>mp3 unit not required for percentage in mass | 4 |
| | (ii) | 0.50 (au); | | 1 |
| | (iii) | $\frac{104 \times 7}{100}$;<br>= 7.28 g;<br>=7.3; | Allow 4/100 for 1 mark<br>or<br>0.28 | 3 |
| (b) | | • change in mass is -1% to +4%<br>• 5% range in mass<br>• between 0.25 and 0.5 au;<br>• accept any answer between 0.375- 0.5;<br><br>***an answer of o.45 scores =4 marks*** | MP 1-4 are separate MPs unless the answer is 0.45 | 4 |
| (c) (i) | | • dry the cylinder/ remove water/liquid;<br>• affects mass/wrong mass/heavier/inaccurate mass/eq; | Accept weight | 2 |
| | (ii) | any two from<br>• volume of (sugar) solution;<br>• temperature;<br>• surface area of cylinders;<br>• same type of potato; | Ignore amount<br>accept volume of liquid | 2<br><br>**Total =16** |

## Type MS · Construct or complete a results or frequency table

### Answer 1 of 6 · Question 4 part a.i · 4 marks

2025 JUN · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |

## Type MS · Construct or complete a results or frequency table

### Answer 2 of 6 · Question 1 part a.i · 3 marks

2021 NOV · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

| Question number | | | | |
|---|---|---|---|---|
| 1 | (a) (i) | <table><tr><th>Conditions</th><th>Result/no/scale of bubbles</th></tr><tr><td>Room temperature</td><td>5</td></tr><tr><td>Alkali added</td><td>3</td></tr><tr><td>Acid added</td><td>2</td></tr><tr><td>3°C</td><td>1</td></tr><tr><td>70°C</td><td>0</td></tr><tr><td>no hydrogen peroxide added</td><td>0</td></tr></table><br>• two column table;<br>• headings;<br>• correct order of results (high to low/ORA); | | 3 marks |
| | (ii) | • no repeats/only done once; | | 1 mark |
| | (iii) | • results based on judgement/subjective/qualitative; | | 1 mark |
| | (iv) | • repeats;<br>• measure volume of oxygen;<br>• using gas syringe/other suitable apparatus; | | 3 marks |
| | (b) (i) | untreated/room temperature; | | 1 mark |
| | (ii) | control; | | 1 mark |
| | (iii) | • reduced/low enzyme activity;<br>• not at optimum pH;<br>• active site changed;<br>• fewer enzyme substrate complexes/substrate won't fit; | | 4 marks |

Total 14 marks

## Type MS · Construct or complete a results or frequency table

### Answer 2 of 6 · Question 1 part a.i · 3 marks

2021 NOV · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Construct or complete a results or frequency table

Answer 3 of 6 · Question 6 part b.ii · 3 marks

2024 NOV · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

<table>
<tr><td>6 (a) (i)</td><td>(cardiac muscle)<br>• striated/striped;<br>• branched;<br>• single/central nucleus;<br>(voluntary muscle)<br>• striated/striped;<br>• unbranched;<br>• many nuclei;<br>(involuntary muscle)<br>• non-striated/striped;<br>• unbranched;<br>• single/central nucleus;<br>• tapered at end;</td><td></td><td>max of 2 for each muscle<br>6</td></tr>
<tr><td></td><td>(ii)</td><td>small intestine/digestive tract/oesophagus/liver/uterus/arteries/veins/iris/sphincter muscles;</td><td>allow any named part of digestive tract</td><td>1</td></tr>
<tr><td>(b) (i)</td><td></td><td>1540;<br>(÷ 20) 77;</td><td>ECF incorrect addition</td><td>2</td></tr>
<tr><td></td><td>(ii)</td><td><table><tr><th>Percentage muscle mass (%)</th><th>Tally/frequency</th></tr><tr><td>70</td><td>ll 2</td></tr><tr><td>72</td><td>ll 2</td></tr><tr><td>74</td><td>llll 4</td></tr><tr><td>75</td><td>II 2</td></tr><tr><td>78</td><td>III 3</td></tr><tr><td>82</td><td>~~llll~~ l 6</td></tr><tr><td>84</td><td>l 1</td></tr></table></td><td>1 mark for suitable table, 1 mark for column headings and 1 mark for correctly inputted data and tally<br>allow just numbers in second column<br>accept just percentage/% as column heading</td><td>3</td></tr>
<tr><td></td><td>(iii)</td><td>82;</td><td>ECF from (ii)</td><td>1</td></tr>
<tr><td>(c)</td><td></td><td>• biceps contract;<br>• triceps relax;<br>• antagonistic pairs/work antagonistically;<br>• lower arm/radius pulled upwards;</td><td></td><td>4</td></tr>
</table>

**Total Marks 17**

## Type MS · Construct or complete a results or frequency table

### Answer 3 of 6 · Question 6 part b.ii · 3 marks

2024 NOV · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |

Type MS · Construct or complete a results or frequency table

Answer 4 of 6 · Question 4 part a.i · 1 mark

2024 JUN · Paper 02 (regional R) · find part a.i in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 4 (a) | (i) | 11, 31, 30 | | (1) |
| | (ii) | 31 (× 100);<br>100<br>= 31%; | send to review if other than 31 | (2) |
| (b) | (i) | **D (alternative forms of the same gene in a genotype)**<br>The following are incorrect because:<br>A two dominant genes are the same type of allele<br>B two recessive genes are the same type of allele<br>C alleles are not different genes | | (1) |
| | (ii) | • sperm and egg/gametes each contain one allele/haploid;<br>• sperm and egg/gametes fuse;<br>• during fertilisation;<br>• (fertilisation) brings alleles together (in diploid cell)/chromosomes (from each gamete) are paired/one allele comes from each parent; | | (4) |
| (c) | (i) | Hh/heterozygous;<br>hh/homozygous recessive; | either order<br>any suitable letters to be used for alleles | (2) |
| | (ii) | wheezing/coughing/shortness of breath/low weight / fatigue/over production of mucus/diarrhoea/male infertility; | allow other valid symptoms | (1) |

**Total for Question 4 = 11 marks**

## Type MS · Construct or complete a results or frequency table

### Answer 5 of 6 · Question 2 part a · 3 marks

2020 JAN · Paper 02 · find part a in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Construct or complete a results or frequency table

Answer 6 of 6 · Question 4 part a.i · 3 marks

2023 JUN · Paper 02 (regional R) · find part a.i in this mark scheme extract

<table>
<tr>
<td>4 (a) (i)</td>
<td>
<table>
<tr><th>Activity</th><th>Pulse rate<br>in beats per minute</th></tr>
<tr><td>walking</td><td></td></tr>
<tr><td>jogging</td><td></td></tr>
<tr><td>running</td><td></td></tr>
</table>
</td>
<td>1 mark for valid table<br>1 mark for labels;<br>1 mark for units;</td>
<td>3</td>
</tr>
<tr>
<td>(ii)</td>
<td>
<table>
<tr><th>Activity</th><th>Pulse rate<br>in beats per minute</th></tr>
<tr><td>walking</td><td>94</td></tr>
<tr><td>jogging</td><td>116</td></tr>
<tr><td>running</td><td>133.5</td></tr>
</table>
</td>
<td>2 marks for all readings correct.<br>1 mark for 1 or 2 readings correct.<br><br>tolerance +/- ½ square</td>
<td>2</td>
</tr>
<tr>
<td>(iii)</td>
<td>Any four from five<br><br>• take resting pulse;<br>• method for measuring pulse rate e.g. finger on wrist/neck/electronic device;<br>• carry out an activity/exercise (for a set period of time/given time);<br>• measure pulse rate after activity;<br>• repeat (for each activity);</td>
<td></td>
<td>4</td>
</tr>
<tr>
<td>(b)</td>
<td>Any four from<br><br>• faster/more/greater muscle contraction during running;<br>• more energy/ATP required;<br>• faster rate of respiration;<br>• more oxygen/ glucose required/more $CO_2$ removed;<br>• increase blood flow/faster blood flow/more blood (to muscles);</td>
<td>ORA for all marking points</td>
<td>4</td>
</tr>
</table>

**Total for Question 4 = 13 marks**